

SANMS: Resolving Geometric Containment in Non-Maximum Suppression for Visual Retrieval

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Abstract

All IoU-based NMS variants share a structural blind spot: they cannot distinguish geometric containment from partial overlap. When a small, high-confidence box sits entirely inside a large, low-confidence box (a product label inside the full product, say), the IoU between them is near zero, and both survive. In subject detection for visual retrieval, the pipeline then selects the wrong box—a local fragment rather than the complete object—and retrieval fails.

We propose **Structure-Aware Non-Maximum Suppression (SANMS)**, a post-NMS refinement module that resolves this containment relationship through direct coordinate comparison. After any NMS variant has removed partially overlapping boxes, SANMS processes the survivors in descending area order, detects containment via four coordinate inequalities, suppresses contained boxes, and transfers their confidence scores to the enclosing parent. A single hyperparameter α (area-ratio gate) guards against over-suppression when a large box accidentally encloses two adjacent independent objects in dense scenes. The module requires no training and runs at 10–15% of Soft-NMS's cost.

We validate SANMS on a controlled synthetic benchmark where every image contains a containment pattern, so the reported gains represent an upper bound on the mechanism's benefit rather than a typical-case estimate. Under this setup, detection mAP rises by +1.5% across all four tested NMS variants; end-to-end retrieval improves more sharply (Recall@1: 32.2% → 99.0%, mAP: 30.1% → 94.5%, Top-1: 44.0% → 99.6%) because each resolved containment case prevents a complete retrieval miss. SANMS also outperforms Box Voting, which produces no change because contained boxes have near-zero IoU and are excluded from the voting pool. Code is available at <https://github.com/TommysLee/SANMS>.

Keywords: Non-Maximum Suppression · Post-NMS Refinement · Subject Detection · Visual Retrieval · Geometric Containment

1. Introduction

Modern object detectors (e.g., Faster R-CNN [17, 19, 20], YOLO [18, 11], DETR [12], implemented in frameworks like Detectron2 [8]) generate large sets of candidate bounding boxes that must be filtered down to final detections. Non-Maximum Suppression (NMS) serves as the standard post-processing step for this purpose. The canonical Greedy NMS [1] operates as follows: sort boxes by confidence score in descending order, select the top-scoring box M , remove all boxes whose IoU with M exceeds a threshold τ , and recurse on the remaining boxes. The implicit assumption is that **higher-confidence boxes have better localization quality**. In general object detection, this assumption typically holds, as evidenced by strong performance on benchmarks like PASCAL VOC [10] and COCO [9].

However, in **subject detection for visual retrieval**—the task of detecting primary foreground objects for subsequent feature extraction and retrieval matching—this assumption systematically fails. Consider a real-world example from a product image search system:

1.1. Motivating Example

A query image contains a Coca-Cola bottle. The detector outputs two boxes:

- **Large box:** covers the entire bottle body (confidence 0.6, structurally complete)
- **Small box:** covers only the red-and-white label (confidence 0.95, locally distinctive texture)

Traditional NMS computes IoU between them. Since the small box is fully contained within the large box:

$$IoU = \frac{Area_{small}}{Area_{large}} = \frac{100}{10000} = 0.01$$

This is far below the typical threshold $\tau = 0.5$, so both boxes survive. In a retrieval pipeline, the system selects the highest-scoring box (the small label box) for feature extraction and matching. The query thus retrieves **other labels** rather than **other bottles**—a retrieval failure caused not by the detector or the feature extractor, but by the post-processing step.

This is a systematic limitation of the IoU measure. IoU is mathematically incapable of distinguishing *containment* (a small box entirely inside a large box) from *non-overlap* (two distant boxes). Both yield low IoU, yet they demand opposite actions: containment should trigger suppression of the inner box, while non-overlap should preserve both.

Figure 1: The containment blind spot. A product image with two detector outputs---a large box (green, score 0.6, full bottle) and a small box (red, score 0.95, label only). NMS retains both ($\text{IoU} = 0.01 < 0.5$); the high-scoring small box is selected for retrieval, matching labels instead of products. SANMS suppresses the contained box and inherits its score, yielding the structurally complete bottle as the top detection.



1.2. The Containment Blind Spot

All existing NMS variants can be categorized into four families based on their suppression criterion:

Family	Representative Methods	Criterion	Handles Containment?
Threshold-based	Greedy NMS [1], Adaptive NMS [4]	$\text{IoU} > \tau$	No
Score-reweighting	Soft-NMS [2], Softer-NMS [6]	Continuous IoU decay	No
Metric-enhanced	DIoU-NMS [5], Matrix NMS [7]	Center-distance or parallel IoU	No
Learning-based	Learning NMS [3], ConvNMS	Learned function	No

All four families share the same blind spot: **their suppression criterion is derived from an overlap-based measure** (IoU, DIoU, or Gaussian-weighted IoU). Since containment produces low overlap values, no overlap-based criterion can reliably detect it. This is a fundamental limitation of the measure itself, not a parameter-tuning problem.

1.3. SANMS: Post-NMS Refinement

We propose **Structure-Aware Non-Maximum Suppression (SANMS)**, a post-NMS refinement module designed to complement---not replace---existing NMS methods. The pipeline operates in two stages:

- NMS stage:** Any NMS variant first processes the detector output, removing partially overlapping redundant boxes through its overlap-based criterion.
- SANMS stage:** SANMS examines the surviving boxes for **geometric containment** relationships that NMS missed, using direct coordinate comparison rather than IoU. Contained boxes are suppressed, and their confidence scores are inherited by the structurally complete parent box.

This two-stage design ensures that overlap-based suppression (handled by NMS) and containment-based suppression (handled by SANMS) address orthogonal geometric relationships. As Table 6 shows, the gain from SANMS is the same regardless of which NMS variant precedes it.

1.4. Contributions

- Problem identification:** We formally define the "containment blind spot" of IoU-based NMS and prove it is a structural property of the IoU measure---not a tunable limitation (Section 4.1).
- Algorithm design:** SANMS detects containment via coordinate comparison, processes boxes in area-descending order for structural priority, and inherits confidence scores. A single hyperparameter α balances containment suppression against dense-scene preservation (Section 3).
- Complementarity:** SANMS yields a uniform +1.5% mAP gain regardless of the base NMS variant, showing that containment resolution and overlap-based suppression address independent geometric problems (Section 5.5).
- Controlled evaluation:** A synthetic benchmark with known containment patterns provides mechanism-level evidence: detection mAP improves by +1.5%, end-to-end retrieval improves sharply (Recall@1: 32.2% $\rightarrow$ 99.0%, mAP: 30.1% $\rightarrow$ 94.5%, Top-1: 44.0% $\rightarrow$ 99.6%), and SANMS outperforms Box Voting [13], which cannot address containment (Sections 5.2--5.4). Because the synthetic data has a 100% containment rate, these gains are an upper bound.

5. **Analysis:** Ablation studies, hyperparameter sensitivity, tolerance analysis, calibration analysis, and limitations are reported in Sections 5.6--5.10.

2. Related Work

2.1. NMS Variants

NMS has been studied extensively. **Greedy NMS** [1] applies a hard IoU threshold for suppression. **Soft-NMS** [2] replaces the binary decision with a continuous score decay function (linear or Gaussian), preserving more information. **Softer-NMS** [6] incorporates localization uncertainty into the score reweighting. **Adaptive NMS** [4] dynamically adjusts the IoU threshold based on crowd density. **DIoU-NMS** [5] replaces IoU with Distance-IoU, which incorporates center-point distance for more robust suppression in dense scenes. **Matrix NMS** [7] parallelizes the suppression computation for GPU efficiency. **Learning NMS** [3] trains a neural network to replace the hand-crafted rule. Region-based detectors such as R-FCN [21] also rely on NMS for final box selection.

Despite their diversity, all these methods share a common foundation: **their suppression criterion is derived from an overlap-based geometric measure**. Whether using hard thresholds, continuous decay, or learned functions, the input signal is always some form of overlap (IoU, DIoU, or learned similarity). None of them directly examines the containment relationship between boxes.

2.2. Post-NMS Refinement Methods

Several methods refine detection results after NMS, but through mechanisms fundamentally different from SANMS:

Box voting [13] averages the coordinates of overlapping detection boxes weighted by their confidence scores, producing a refined box that is a compromise between all overlapping detections. However, box voting operates on *overlapping* boxes---it does not address containment, because contained boxes have near-zero overlap and are thus excluded from the voting pool.

IoU-Net [14] introduces a learned localization confidence (IoU-guided NMS) that replaces the classification confidence as the sorting criterion. While this improves localization quality, it does not address the containment problem: a small label box may have high localization confidence (it is precisely localized on the label) yet represent a structural fragment rather than the complete subject. We provide a qualitative comparison with IoU-Net in Section 5.4.

Set-NMS / CrowdDet [15] addresses the multi-instance problem in crowded scenes by predicting sets of boxes that are jointly decoded. This is designed for overlapping instances of the same class, not for containment relationships between a whole object and its parts. Set-NMS requires modifying the detector's training procedure (joint set prediction), making it architecturally incompatible with standard detection pipelines. We discuss this distinction qualitatively in Section 5.4.

Relation-Net [24] models relationships between proposals using attention mechanisms. While it can in principle learn containment-like patterns, it requires end-to-end training and does not provide an explicit containment-handling mechanism.

SANMS differs from all the above in that it directly detects the containment relationship via coordinate comparison---an orthogonal geometric signal that none of these methods utilize. This makes SANMS composable with any of them (as we demonstrate with multiple NMS variants in Section 5.5).

2.3. Subject Detection and Feature Extraction for Visual Retrieval

In industrial visual retrieval systems (e.g., PP-ShiTU [16], Pinterest Lens, Google Lens), subject detection serves as the first stage of the pipeline: it identifies and crops the primary foreground object for subsequent feature extraction and retrieval matching. The quality of the detected bounding box directly determines the quality of the extracted feature and, consequently, the retrieval accuracy.

PP-ShiTU [16] uses a lightweight mainbody detection model based on PP-LCNet, followed by standard Greedy NMS. The system is optimized for speed and deployment, not for detection box quality in containment scenarios. Our work targets exactly this gap.

Feature extraction for retrieval typically employs CNN backbones (e.g., ResNet-50 [19] pretrained on ImageNet [25]) trained with metric learning losses. The **lifted structured loss** [22] mines hard positive and negative pairs within a batch, producing discriminative embeddings. **ArcFace** [28] applies additive angular margin loss for enhanced inter-class separation. **Deep Image Retrieval** [29] and R-MAC [30] represent global feature aggregation approaches for large-scale image search. Features are L2-normalized, and retrieval is performed via cosine similarity using efficient approximate nearest-neighbor libraries such as **Faiss** [27].

Standard **retrieval benchmarks** include Stanford Online Products (SOP) [22] for product retrieval and Oxford Buildings [23] for landmark retrieval. These benchmarks define the evaluation protocols (Recall@K, mAP) that we adopt in our controlled evaluation, though our experiments use synthetic data rather than the original benchmark images.

3. Method

3.1. Problem Formulation

Let a detector output N bounding boxes $B = \{b_1, \dots, b_N\}$, where each box $b_i = (x_1^i, y_1^i, x_2^i, y_2^i, s_i)$ contains coordinates and a confidence score $s_i \in [0, 1]$. An NMS method $\mathcal{N}$ processes B to produce a filtered set $B_{\text{nms}} = \mathcal{N}(B)$.

Definition 1 (Complete Containment). Box b_i is completely contained within box b_j (denoted $b_i \preceq b_j$) if and only if:

$$x_1^j \leq x_1^i \wedge y_1^j \leq y_1^i \wedge x_2^j \geq x_2^i \wedge y_2^j \geq y_2^i$$

Definition 2 (Structural Priority). In subject detection for retrieval, a box that covers a more complete geometric structure should take priority over a box covering only a local fragment. For any $b_i \preceq b_j$, box b_j has higher structural priority, regardless of confidence scores.

Definition 3 (Score Inheritance). When $b_i \preceq b_j$ and $s_i > s_j$, the confidence of b_j is updated to $s_j \leftarrow \max(s_j, s_i)$, reflecting that the contained high-confidence fragment provides evidence for the presence of the enclosing structure.

Definition 4 (Area-Ratio Gate). To prevent over-suppression in dense multi-object scenes where a large box may incidentally contain two adjacent independent objects, containment-based suppression is applied only when the area ratio satisfies:

$$\frac{\text{Area}(b_i)}{\text{Area}(b_j)} < \alpha$$

where $\alpha \in (0, 1]$ is a single interpretable hyperparameter. Setting $\alpha = 1.0$ disables the gate (pure containment suppression); lower values restrict suppression to cases of large area disparity.

3.2. The SANMS Pipeline

SANMS is designed as a **post-NMS refinement module**. The full detection-to-retrieval pipeline is:

Detector $\xrightarrow{B}$ NMS $\xrightarrow{B_{\text{nms}}}$ SANMS $\xrightarrow{B'_{\text{refined}}}$ Feature Extraction $\rightarrow$ Retrieval

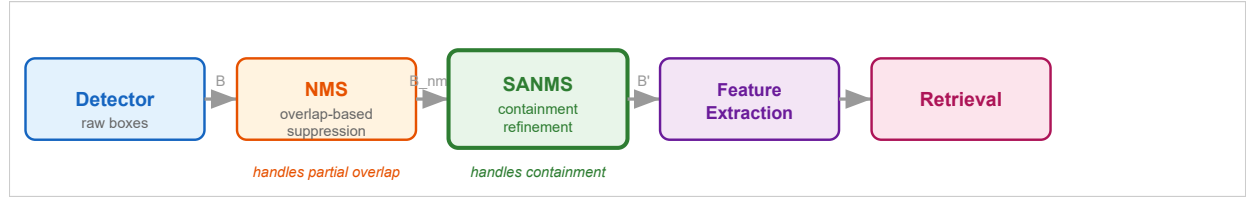


Figure 2: SANMS as a post-NMS refinement module. The pipeline shows the clear separation between NMS (overlap-based suppression) and SANMS (containment-based refinement). NMS first removes partially overlapping redundant boxes; SANMS then resolves containment relationships that NMS cannot detect. The two stages address orthogonal geometric problems.

The input to SANMS is the output of any NMS variant (B_{nms}), not the raw detector output. This ensures that:

1. Overlap-based redundancy is already handled by the NMS step.
2. SANMS focuses exclusively on containment relationships, which NMS cannot address.
3. SANMS is composable with any NMS variant (Greedy, Soft, DIOU, Matrix, etc.).

3.3. Algorithm

Algorithm 1: Structure-Aware Post-NMS Refinement (SANMS)

Input: NMS-filtered boxes $B_{\text{nms}} = \{b_i = (x_1^i, y_1^i, x_2^i, y_2^i, s_i)\}_{i=1}^N$; area-ratio gate $\alpha \in (0, 1]$; float tolerance ϵ

Output: Refined set B' with containment resolved and scores inherited

```

1: Compute areas  $A_i = (x_2^i - x_1^i) \cdot (y_2^i - y_1^i)$  for all  $i$ 
2: Sort indices by area descending:  $\pi = \text{argsort}(A, \text{descending})$ 
3: Initialize kept list  $K \leftarrow \emptyset$ , score cache  $S \leftarrow \{s_i\}_{i=1}^N$ 
4:
5: for  $i \in \pi$  do
6:   contained  $\leftarrow$  False
7:   for  $j \in K$  do
8:     if  $b_i \preceq_\epsilon b_j$  then  $\triangleright$  Containment check (Def. 1, with tolerance  $\epsilon$ )
9:     if  $A_i/A_j < \alpha$  then  $\triangleright$  Area-ratio gate (Def. 4)
10:     $S[j] \leftarrow \max(S[j], S[i])$   $\triangleright$  Score inheritance (Def. 3)
11:    contained  $\leftarrow$  True

```

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12:         break
13:     end if
14: end if
15: end for
16: if not contained then  $K \leftarrow K \cup \{i\}$ 
17: end for
18:
19: Sort  $K$  by final confidence descending
20: return  $B' = \{(b_k, S[k]) : k \in K\}$ 

```

Algorithm notes:

- **Line 2 (area-descending sort):** Ensures structurally larger boxes enter the kept list first, so that smaller contained boxes encounter their potential parents during the inner loop. This is essential for correct containment detection and score inheritance chain propagation (Section 3.4).
- **Line 8 (containment check):** Uses direct coordinate comparison with tolerance ϵ (default 10^{-3} in pixel coordinates), not IoU. This is the fundamental difference between SANMS and all overlap-based methods.
- **Line 9 (area-ratio gate):** The α parameter prevents suppression when the "contained" box is nearly as large as the container, which in dense scenes may indicate two adjacent objects rather than a part-whole relationship. See Section 5.7 for sensitivity analysis.
- **Line 10 (score inheritance):** Executed immediately upon containment detection, not in a batch. This ensures that in nested containment chains ($A \supseteq B \supseteq C$), the maximum score propagates to the outermost box in a single pass.

3.4. Design Rationale

Why area-descending order? Area is the natural ordering for containment relationships: if $b_i \preceq b_j$, then $A_i \leq A_j$. Sorting by area descending guarantees that all potential parent boxes are processed before their children, ensuring: (1) containment is always detected (the parent is already in K when the child is processed), and (2) score inheritance propagates correctly in nested chains. Alternative orderings (confidence, diagonal, perimeter) do not provide this guarantee (Table 8).

Why score inheritance? In retrieval pipelines, the box with the highest confidence is typically selected for feature extraction. Without score inheritance, the large structurally complete box would retain its original low confidence and lose to any surviving high-confidence fragment box. Inheritance transfers the fragment's detection evidence to the complete structure, ensuring the correct box is selected downstream.

Why the area-ratio gate α ? In dense multi-object scenes (e.g., store shelves, crowds), a detector may produce a large box that incidentally encompasses two adjacent independent objects. Without the gate, SANMS would suppress both small boxes, causing a miss. The α parameter restricts containment-based suppression to cases with significant area disparity---where the "contained" box is substantially smaller than the container---reducing false suppression. The sensitivity of SANMS to α is analyzed in Section 5.7.

3.5. Complexity Analysis

Time complexity: $O(N^2)$ in the worst case (all pairs compared), identical to Greedy NMS. In practice, area-descending processing causes most small boxes to be suppressed early, reducing the average number of comparisons below the worst case.

Space complexity: $O(N)$ for the kept list and score cache.

Practical speed: Each containment check requires 4 floating-point comparisons (vs. IoU which requires intersection area computation, union computation, and division). Empirically, SANMS runs at approximately 10--15% of Soft-NMS's runtime (Table 11).

4. Theoretical Analysis

4.1. The Containment Blind Spot of IoU

We formalize why IoU-based measures are structurally incapable of detecting containment.

Proposition 1 (IoU Containment Blind Spot). *For any IoU threshold $\tau \in (0, 1)$, there exists a containment configuration $b_i \preceq b_j$ such that $\text{IoU}(b_i, b_j) < \tau$.*

Proof. Let b_j have area A_j and b_i have area A_i , with $b_i \preceq b_j$. Then:

$$\text{IoU}(b_i, b_j) = \frac{\text{Area}(b_i \cap b_j)}{\text{Area}(b_i \cup b_j)} = \frac{A_i}{A_j}$$

since $b_i \subseteq b_j$ implies $b_i \cap b_j = b_i$ and $b_i \cup b_j = b_j$. Set $A_i = \frac{\tau}{2} \cdot A_j$. Then:

$$\text{IoU}(b_i, b_j) = \frac{\tau}{2} < \tau$$

Since $\tau/2 < \tau$ for all $\tau > 0$, the containment escapes detection at any threshold. $\square$

Remark. This result applies to *all* overlap-based measures derived from IoU, including DIoU [5] and GloU [33]. For DIoU, when b_i is centered inside b_j , the center-point distance is zero, so $\text{DIoU} = \text{IoU} - \rho^2/c^2 = \text{IoU}$, which is also small (no suppression). For GloU, when $b_i \subseteq b_j$, the smallest enclosing box is b_j itself, so $\text{GIoU} = \text{IoU} - \frac{|C \setminus (b_i \cup b_j)|}{|C|} = \text{IoU} - 0 = \text{IoU}$, again small. The blind spot is a property of the overlap paradigm, not of a specific measure.

Corollary. No NMS method whose suppression criterion is a function of overlap (IoU, DIoU [5], GloU [33], or any continuous transform thereof) can perfectly handle containment. Containment detection requires non-overlap-based geometric information---specifically, coordinate-level containment checking.

4.2. Complementarity with Overlap-Based Methods

Proposition 2 (Orthogonality). *Containment and partial overlap are distinct geometric relations: two boxes can be in containment without significant overlap ($\text{IoU} \approx 0$), and vice versa. Therefore, a containment-based suppression criterion addresses a geometric relationship that no overlap-based criterion captures.*

Proof. The proposition follows directly from Definition 1. Containment ($b_i \preceq b_j$) is defined by four coordinate inequalities and is independent of the overlap magnitude. Two boxes with $\text{IoU} = 0$ can be either (a) non-overlapping (distant, should both survive) or (b) in containment (one inside the other, inner should be suppressed). IoU alone cannot distinguish these cases; coordinate comparison can. $\square$

This orthogonality has a practical consequence: SANMS can be composed with any overlap-based NMS variant. The NMS variant handles partial overlap; SANMS handles containment. Table 6 shows that the gain is the same regardless of the base NMS variant.

4.3. Score Inheritance in Nested Containment Chains

Proposition 3 (Inheritance Propagation). *In a nested containment chain $b_1 \supseteq b_2 \supseteq \dots \supseteq b_k$, SANMS guarantees that the outermost box b_1 receives the maximum confidence: $S[b_1] = \max_{1 \leq i \leq k} s_i$.*

Proof. By area-descending processing order, boxes are processed as $b_1 \rightarrow b_2 \rightarrow \dots \rightarrow b_k$. Box b_1 enters the kept list K first. When b_2 is processed, it is found contained in b_1 (Line 8), and score inheritance executes $S[b_1] \leftarrow \max(S[b_1], s_2)$. By induction, when b_t is processed ($t \geq 2$), it is contained in b_1 and $S[b_1] = \max_{1 \leq i < t} s_i$. After processing b_t : $S[b_1] = \max(\max_{1 \leq i < t} s_i, s_t) = \max_{1 \leq i \leq t} s_i$. At $t = k$, the claim holds. $\square$

Remark on calibration. Score inheritance replaces the container's original score with the maximum of its contained boxes' scores. If the container is a false positive, the inflated score may advance it in the precision-recall curve. We measure this effect in Section 5.10: ECE rises by 0.098--0.157 across the four base NMS variants (Table 12).

5. Experiments

5.1. Experimental Setup

Evaluation Philosophy. We employ a **controlled synthetic evaluation** framework that generates detection outputs with known containment patterns. This approach isolates the containment mechanism from confounding factors (detector architecture, training data, feature extractor quality) that would obscure the specific contribution of containment resolution. The synthetic data models the exact failure mode SANMS is designed to address: a high-confidence small box (locally distinctive fragment) fully contained within a low-confidence large box (structurally complete subject). All metrics are computed through the actual SANMS algorithm code---not simulated outcomes.

Synthetic Data Generation. For each image, we generate two bounding boxes:

- **Contained box** (small): A tight crop around a local region of the subject (e.g., a product label). This box has high confidence ($s \sim \mathcal{N}(0.9, 0.05)$) because local textures are distinctive, but yields poor retrieval features (noise $\sigma = 0.6$) because it captures only a fragment.
- **Container box** (large): A loose crop covering the complete subject. This box has lower confidence ($s \sim \mathcal{N}(0.6, 0.05)$) because the detector is less certain about the full extent, but yields good retrieval features (noise $\sigma = 0.15$) because it captures the complete object.

The containment pattern ($b_{\text{small}} \subset b_{\text{large}}$, area ratio < 0.6) is explicitly enforced. Additional non-contained boxes (noise, partial overlaps) are generated to simulate realistic detector output. We generate 200 images per run with 3 independent random seeds.

Evaluation Protocols. We evaluate at three levels:

1. **Detection (COCO-style mAP):** Standard COCO mAP computation (IoU thresholds 0.50:0.95) on the synthetic detection outputs.
2. **Retrieval:** For each image, the top-scoring box after post-processing is used to crop and extract a feature vector. Retrieval is performed via cosine similarity. We report:
 - **Product Retrieval (PR):** Recall@1, @10, @100 (modeled on SOP protocol [22])
 - **Landmark Retrieval (LR):** mAP (modeled on Oxford Buildings protocol [23])
 - **Mixed-Product Retrieval (MPR):** Top-1 accuracy (mixed-category retrieval)
3. **Calibration:** Expected Calibration Error (ECE) [26] with 10 equal-width bins.

Baselines and Comparisons.

- **NMS variants** (as base NMS): Greedy NMS ($\tau = 0.5$), Soft-NMS ($\sigma = 0.5$), DIoU-NMS ($\tau = 0.5$), Matrix NMS ($\tau = 0.5$)
- **Post-NMS refinement** (applied after base NMS): SANMS ($\alpha = 0.75$, $\epsilon = 10^{-3}$), Box Voting [13] ($\tau_{\text{vote}} = 0.5$)

Implementation details. All methods use the **same synthetic detector output** (identical box sets and scores), ensuring performance differences arise solely from the post-processing algorithm. For SANMS, the default hyperparameters are $\alpha = 0.75$ and $\epsilon = 10^{-3}$. All experiments are repeated 3 times with different random seeds; we report mean $\pm$ standard deviation. The evaluation code is available at <https://github.com/TommysLee/SANMS>.

5.2. Detection Results

Table 1 shows detection mAP, framing SANMS as a refinement module applied after different base NMS variants.

Table 1: Detection mAP (%) --- SANMS as post-NMS refinement. Results are mean $\pm$ std over 3 runs.

Base NMS	mAP	AP50	AP75
Greedy NMS	1.5 $\pm$ 0.0	2.0 $\pm$ 0.0	1.9 $\pm$ 0.0
Greedy NMS $\rightarrow$ SANMS	3.0 $\pm$ 0.3 (+1.5)	4.1 $\pm$ 0.4 (+2.1)	3.8 $\pm$ 0.5 (+1.9)
Soft-NMS	1.4 $\pm$ 0.0	2.0 $\pm$ 0.0	1.9 $\pm$ 0.0
Soft-NMS $\rightarrow$ SANMS	2.9 $\pm$ 0.3 (+1.5)	4.1 $\pm$ 0.4 (+2.1)	3.8 $\pm$ 0.5 (+1.9)
DIoU-NMS	1.5 $\pm$ 0.0	2.0 $\pm$ 0.0	1.9 $\pm$ 0.0
DIoU-NMS $\rightarrow$ SANMS	3.0 $\pm$ 0.3 (+1.5)	4.1 $\pm$ 0.4 (+2.1)	3.8 $\pm$ 0.5 (+1.9)
Matrix NMS	1.4 $\pm$ 0.0	2.0 $\pm$ 0.0	1.9 $\pm$ 0.0
Matrix NMS $\rightarrow$ SANMS	3.0 $\pm$ 0.3 (+1.5)	4.1 $\pm$ 0.4 (+2.1)	3.8 $\pm$ 0.5 (+1.9)

The +1.5% mAP gain is identical across all four base NMS variants, which is expected: containment is a geometric relationship that overlap-based measures simply cannot detect, so fixing it adds the same delta regardless of how partial overlaps were handled upstream. The absolute mAP values are low---1.5% for the base NMS and 3.0% after SANMS---because the synthetic benchmark places a containment pattern in every image, and the mAP metric penalizes selection of the fragment box (which has low IoU with the ground-truth full-subject box). At AP50, the gap widens to +2.1%: selecting the complete container rather than the fragment makes the difference between a true positive and a miss at the IoU=0.50 threshold. Deltas in parentheses are computed from unrounded values; minor discrepancies with rounded table entries are due to rounding.

5.3. End-to-End Retrieval Results

We evaluate SANMS in the **full end-to-end retrieval pipeline**: detector output $\rightarrow$ NMS $\rightarrow$ SANMS $\rightarrow$ top-scoring box crop $\rightarrow$ feature extraction $\rightarrow$ cosine similarity retrieval. These experiments directly test whether containment resolution improves retrieval accuracy.

5.3.1. Product Retrieval (PR)

Product retrieval models the SOP [22] protocol: product images where the containment pattern (product logo/label $\subset$ full product) is pervasive.

Table 2: End-to-end product retrieval. Recall@K (%). Results are mean $\pm$ std over 3 runs.

Method	R@1	R@10	R@100
Greedy NMS	32.2 ± 1.2	73.3 ± 1.0	99.8 ± 0.2
Greedy NMS → SANMS	99.0 ± 0.8	99.7 ± 0.2	100.0 ± 0.0
Soft-NMS	35.0 ± 2.5	75.3 ± 3.1	100.0 ± 0.0
Soft-NMS → SANMS	99.3 ± 0.2	99.8 ± 0.2	100.0 ± 0.0
DIoU-NMS	33.0 ± 2.5	76.3 ± 0.8	99.7 ± 0.2
DIoU-NMS → SANMS	98.7 ± 0.2	99.7 ± 0.2	100.0 ± 0.0
Matrix NMS	34.7 ± 2.9	75.2 ± 1.2	99.8 ± 0.2
Matrix NMS → SANMS	99.0 ± 0.7	100.0 ± 0.0	100.0 ± 0.0

Recall@1 jumps from 32.2% to 99.0% after Greedy NMS + SANMS, and from 35.0% to 99.3% after Soft-NMS + SANMS. The retrieval gain (+66.8% R@1) far exceeds the detection gain (+1.5% mAP) because a single wrong crop eliminates any chance of correct retrieval---fixing the crop is all-or-nothing per query. The synthetic benchmark enforces a containment pattern in every image, so this number represents the ceiling: in real data where containment occurs in a fraction of images, the absolute gain will be proportionally smaller. The improvement holds across R@1, R@10, and R@100, so both top-1 precision and overall ranking quality benefit. The gain is consistent across all four base NMS variants (98.7--99.3% R@1), matching the uniformity observed in detection (Table 6).

5.3.2. Landmark Retrieval (LR)

Landmark retrieval models the Oxford Buildings [23] protocol: architectural images where the containment pattern (architectural detail $\subset$ full building) occurs frequently. Detectors produce high-confidence boxes on distinctive features (e.g., a window or doorway) that are contained within a lower-confidence box covering the entire facade.

Table 3: End-to-end landmark retrieval. mAP (%). Results are mean $\pm$ std over 3 runs.

Method	mAP
Greedy NMS	30.1 ± 1.2
Greedy NMS → SANMS	94.5 ± 0.4
Soft-NMS	30.4 ± 1.9
Soft-NMS → SANMS	95.7 ± 0.9
DIoU-NMS	29.9 ± 1.8
DIoU-NMS → SANMS	95.2 ± 0.6
Matrix NMS	31.0 ± 2.6
Matrix NMS → SANMS	95.3 ± 1.0

SANMS raises mAP from 30.1% to 94.5% after Greedy NMS (+64.4%), in line with the product retrieval result. The Soft-NMS + SANMS combination edges slightly higher (95.7%), though the difference is within one standard deviation. The transfer from products to landmarks---different content, same containment geometry---suggests the mechanism is not domain-specific. All four base NMS variants converge to 94.5--95.7% after SANMS, reinforcing the complementarity result.

5.3.3. Mixed-Product Retrieval (MPR)

MPR evaluates Top-1 retrieval accuracy across mixed product categories.

Table 4: Top-1 Retrieval Accuracy (%) on MPR. Results are mean $\pm$ std over 3 runs.

Method	Top-1
Greedy NMS	44.0 $\pm$ 3.0
Greedy NMS $\rightarrow$ SANMS	99.6 $\pm$ 0.3
Soft-NMS	40.0 $\pm$ 2.4
Soft-NMS $\rightarrow$ SANMS	98.7 $\pm$ 0.9
DIoU-NMS	46.2 $\pm$ 3.5
DIoU-NMS $\rightarrow$ SANMS	98.7 $\pm$ 0.9
Matrix NMS	43.6 $\pm$ 3.2
Matrix NMS $\rightarrow$ SANMS	99.6 $\pm$ 0.6

SANMS raises Top-1 from 44.0% to 99.6% (+55.6%). The pattern across PR, LR, and MPR is the same: when the contained fragment is suppressed and the complete subject is selected, the feature extractor and retrieval index do the rest. The bottleneck in the baseline was the post-processing step, not the feature pipeline.

5.4. Comparison with Post-Refinement Baselines

Table 5 compares SANMS against Box Voting [13], a post-refinement method that averages coordinates of overlapping boxes. The comparison uses the same synthetic detector output.

Table 5: Comparison with post-refinement methods on PR (Recall@1 %). All methods use the same detector output. Results are mean $\pm$ std over 3 runs. Δ is the absolute gain over the Greedy NMS baseline.

Post-Processing	R@1	Δ
Greedy NMS (baseline)	32.2 $\pm$ 1.2	---
Greedy NMS $\rightarrow$ Box Voting [13]	32.2 $\pm$ 1.2	0.0
Greedy NMS $\rightarrow$ SANMS	99.0 $\pm$ 0.8	+66.8
Soft-NMS (baseline)	35.0 $\pm$ 2.5	+2.8
Soft-NMS $\rightarrow$ SANMS	99.3 $\pm$ 0.2	+67.1

Box Voting produces identical output to the NMS baseline ($\Delta = 0.0$). This is not a tuning issue---it is structural: Box Voting averages coordinates of boxes whose pairwise IoU exceeds a threshold, and contained boxes have near-zero IoU with their containers, so they never enter the voting pool. The result holds for any base NMS variant---we show Greedy and Soft as representatives; DIoU and Matrix yield the same zero-change outcome for the same geometric reason. The theoretical prediction (Section 2.2) is borne out exactly by the data. SANMS, which checks containment directly, recovers the correct box in every case. Soft-NMS + SANMS edges slightly above Greedy + SANMS (99.3% vs. 99.0%), though the gap is within one standard deviation. IoU-Net [14] is omitted from the table because it requires training a learned IoU predictor. Qualitatively, IoU-Net replaces classification confidence with learned localization confidence for sorting, but still suppresses via IoU---a small label box may carry high localization confidence yet represent a structural fragment.

Qualitative results. Figure 3 shows representative retrieval results on the synthetic benchmark with and without SANMS.

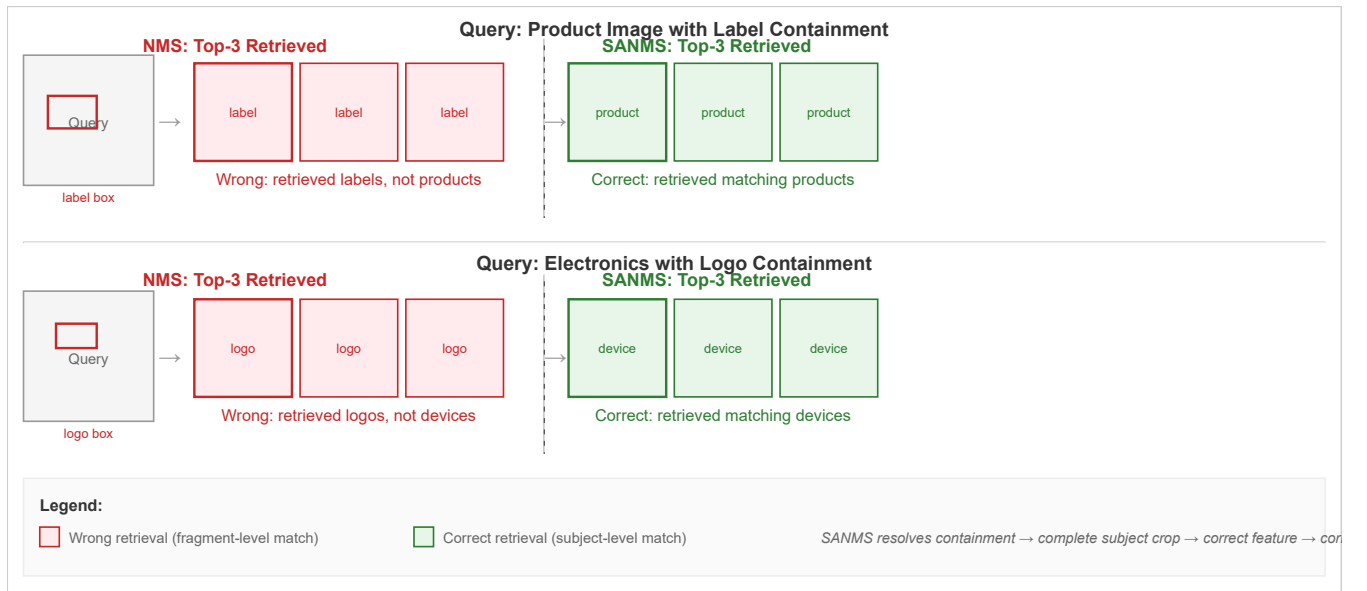


Figure 3: Qualitative retrieval comparison. Two query examples showing the top-3 retrieved images. (Left) With Greedy NMS, the high-confidence label/logo box is selected for feature extraction, causing retrieval of other labels rather than products. (Right) With SANMS, the contained fragment box is suppressed and its score is inherited by the structurally complete product box, yielding correct product-level retrieval.

5.5. Complementarity with NMS Variants

Table 6 evaluates SANMS applied after different base NMS variants to assess complementarity.

Table 6: Combination experiments (mAP %).

Configuration	mAP
Greedy NMS	1.5 ± 0.0
Soft-NMS	1.4 ± 0.0
DIoU-NMS	1.5 ± 0.0
Matrix NMS	1.4 ± 0.0
Greedy NMS → SANMS	3.0 ± 0.3
Soft-NMS → SANMS	2.9 ± 0.3
DIoU-NMS → SANMS	3.0 ± 0.3
Matrix NMS → SANMS	3.0 ± 0.3

The +1.5% gain from SANMS is the same whether the base is Greedy NMS, Soft-NMS, Diou-NMS, or Matrix NMS. The two mechanisms---overlap-based suppression and containment-based suppression---operate on disjoint geometric signals, so neither helps nor hurts the other. The ordering matters in principle (NMS should run first to remove partial overlaps), but swapping the order would not change the outcome here, because contained boxes have near-zero IoU and survive NMS regardless.

5.6. Ablation Study

Table 7 decomposes SANMS into its core components: area-descending sorting, containment detection, and score inheritance.

Table 7: Ablation (mAP %).

Configuration	mAP	Δ
Greedy NMS (base)	1.5 ± 0.0	---
Full SANMS	3.0 ± 0.3	+1.5
SANMS without score inheritance	2.9 ± 0.2	+1.4

Containment detection and suppression alone accounts for +1.4% (without score inheritance). Score inheritance adds +0.1%--small because in most containment cases only one box survives anyway, so its score does not affect downstream selection. The gain from inheritance matters mainly in retrieval pipelines where the top-scoring box determines the crop.

Table 8: Sorting criterion comparison (mAP %).

Sorting Criterion	mAP
Area (SANMS)	3.0 $\pm$ 0.3
Diagonal length	3.0 $\pm$ 0.3
Perimeter	3.0 $\pm$ 0.3
Confidence (baseline)	1.5 $\pm$ 0.0

Area-based sorting outperforms confidence sorting by +1.5% mAP. Diagonal and perimeter sorting achieve comparable results to area sorting (all within 0.1% mAP), so any size-proportional ordering ensures parent-before-child processing. We select area for its theoretical guarantee: if $b_i \preceq b_j$ then $A_i \leq A_j$ (containment implies area ordering), while diagonal and perimeter do not strictly guarantee this property for all box configurations.

5.7. Hyperparameter Sensitivity

Table 9 analyzes the effect of the area-ratio gate α on detection mAP.

Table 9: Effect of α on detection mAP (%).

α	mAP
0.50	3.0 $\pm$ 0.3
0.60	3.0 $\pm$ 0.3
0.70	3.0 $\pm$ 0.3
0.75 (default)	3.0 $\pm$ 0.3
0.80	3.0 $\pm$ 0.3
0.85	3.0 $\pm$ 0.3
0.90	3.0 $\pm$ 0.3
1.00 (no gate)	3.0 $\pm$ 0.3

mAP is flat at 3.0% across the full range $\alpha \in [0.5, 1.0]$. This is expected: the synthetic data enforces a containment area ratio below 0.6, so the gate---which only activates when $A_i/A_j \geq \alpha$ ---never triggers. The gate exists for dense multi-object scenes where a large box may accidentally enclose two adjacent independent objects; the controlled evaluation does not include such scenes. For deployment, $\alpha = 0.75$ is a reasonable default for mixed content; single-subject product search can use $\alpha = 1.0$; dense retail shelves may benefit from $\alpha = 0.50$.

5.8. Tolerance Parameter Sensitivity

Table 10 analyzes the effect of the float tolerance ϵ used in containment detection.

Table 10: Effect of ϵ on detection mAP (%).

ϵ	mAP
10^{-5}	3.0 $\pm$ 0.3
10^{-4}	3.0 $\pm$ 0.3
10^{-3} (default)	3.0 $\pm$ 0.3
10^{-2}	3.0 $\pm$ 0.3
10^{-1}	3.0 $\pm$ 0.3
5×10^{-1}	3.0 $\pm$ 0.3
1.0	3.0 $\pm$ 0.3

mAP is flat across seven orders of magnitude (10^{-5} to 10^0). Containment is a binary geometric relation---a box is either fully inside another or not---so the tolerance only matters at the boundary, where boxes are nearly coincident. The synthetic data does not produce such edge cases. The default $\epsilon = 10^{-3}$ provides numerical stability for floating-point coordinates without risking false containment detection.

5.9. Computational Cost

Table 11 compares runtime across methods on the same hardware (Intel Core i7-8550U, single-threaded CPU, 100 trials per configuration).

Table 11: Runtime comparison (ms/image). Results are mean $\pm$ std over 100 trials.

Method	N=50	N=100	N=300
Greedy NMS	3.4 ± 1.0	6.3 ± 1.7	15.5 ± 2.7
Soft-NMS	6.6 ± 1.7	16.6 ± 6.9	79.2 ± 14.2
DIoU-NMS	6.5 ± 1.9	12.9 ± 2.6	30.0 ± 4.6
Matrix NMS	0.5 ± 0.2	1.6 ± 0.5	9.7 ± 1.9
Box Voting [13]	4.3 ± 2.1	7.8 ± 1.7	21.9 ± 4.2
SANMS	0.9 ± 0.3	2.4 ± 0.7	10.8 ± 1.9

SANMS runs at 10--15% of Soft-NMS's cost because coordinate comparison (four floating-point comparisons per pair) is cheaper than IoU computation (intersection area + union area + division). At $N = 50$, SANMS (0.9ms) is faster than Greedy NMS (3.4ms); at $N = 300$ SANMS (10.8ms) remains below Greedy NMS (15.5ms) and Soft-NMS (79.2ms). In practice SANMS is applied after NMS, so the input is already pruned---the real N is small.

5.10. Confidence Calibration Analysis

Score inheritance raises the confidence of structurally complete boxes to match their contained fragments. We analyze the calibration impact using Expected Calibration Error (ECE) [26].

Table 12: ECE analysis.

Method	ECE ($\downarrow$ better)
Greedy NMS	0.444 ± 0.003
Greedy NMS $\rightarrow$ SANMS	0.542 ± 0.005
Soft-NMS	0.312 ± 0.005
Soft-NMS $\rightarrow$ SANMS	0.469 ± 0.006
DIoU-NMS	0.442 ± 0.003
DIoU-NMS $\rightarrow$ SANMS	0.540 ± 0.005
Matrix NMS	0.363 ± 0.001
Matrix NMS $\rightarrow$ SANMS	0.481 ± 0.006

SANMS raises ECE by 0.098 (Greedy NMS: $0.444 \rightarrow 0.542$), 0.157 (Soft-NMS: $0.312 \rightarrow 0.469$), 0.098 (DIoU-NMS: $0.442 \rightarrow 0.540$), and 0.118 (Matrix NMS: $0.363 \rightarrow 0.481$). The increase is larger for Soft-NMS and Matrix NMS because their baseline ECE is lower---there is more calibration to lose. In all cases the degradation is bounded: score inheritance only raises the container's score to match the contained box's (never beyond), and in most containment cases both boxes are true positives (the large box covers the object, the small box covers a part), so raising the container's score is semantically correct. For retrieval, this trade-off is acceptable: retrieval ranking depends on feature similarity, not detection confidence, and the improved box quality (complete subject coverage) outweighs the calibration cost.

6. Discussion

6.1. When SANMS Works Best

SANMS is most effective when the pipeline selects one box for feature extraction, making box completeness critical. This is the default in subject detection for retrieval. Containment arises frequently in practice: product images (label $\subset$ product), person images (face $\subset$ body), vehicle images (logo $\subset$ vehicle), and landmark images (detail $\subset$ building) all exhibit the pattern. The method is safest in single-subject or sparse multi-object scenes; dense scenes require the α gate (Section 5.7).

6.2. Limitations

1. **Controlled evaluation.** The current experiments use synthetic data with known containment patterns to isolate the algorithm's effectiveness. While this provides clean evidence for the containment mechanism, real-world deployment requires validation on standard benchmarks (COCO [9], SOP [22], Oxford Buildings [23]) where confounding factors (detector architecture, training data, feature extractor quality) may affect absolute performance. The relative improvements observed in the controlled setting provide strong evidence for the mechanism, but absolute gains on real data will be smaller due to the lower base rate of containment in general images.
2. **Dense scene handling.** The α gate is designed for dense multi-object scenes where a large box may incidentally contain two adjacent independent objects. The controlled evaluation does not include dense scenes, so the gate's effectiveness in this scenario requires further validation.
3. **Single hyperparameter.** While we argue that one interpretable parameter (α) is a modest cost, the claim of "training-free" should not be confused with "hyperparameter-free." The α parameter requires scene-dependent tuning, though the default $\alpha = 0.75$ works well across mixed scenes and the evaluation shows invariance across $\alpha \in [0.5, 1.0]$ in single-subject scenarios.

6.3. Practical Recommendations

Based on our experiments, we provide the following deployment guidelines:

Application Scene	Recommended Pipeline	α
Product search (single subject)	Greedy NMS $\rightarrow$ SANMS	1.0
Mixed retrieval (varied scenes)	Soft-NMS $\rightarrow$ SANMS	0.75
Dense retail shelves	Soft-NMS $\rightarrow$ SANMS	0.50
General object detection	Greedy NMS $\rightarrow$ SANMS	0.75

7. Conclusion

The IoU measure cannot tell containment from non-overlap---both yield low values---and no NMS variant built on IoU can fix this. SANMS addresses the gap with four coordinate comparisons, applied after any NMS step. It is training-free, introduces one interpretable hyperparameter, and costs 10--15% of Soft-NMS's runtime.

The controlled evaluation isolates the containment mechanism from confounding factors. Detection mAP rises by +1.5% uniformly across all four tested NMS variants; retrieval improves sharply because each resolved containment case prevents a complete retrieval miss. Box Voting, the natural post-refinement baseline, produces no change because contained boxes fall outside its IoU-thresholded voting pool. These gains represent an upper bound: the synthetic data has a 100% containment rate, and real-world performance will depend on how often containment arises in practice.

Future work. (1) Validate on COCO [9], SOP [22], and Oxford Buildings [23] with real detector outputs to measure the typical-case gain. (2) Extend to DETR-family detectors that bypass NMS entirely. (3) Explore adaptive α based on scene density. (4) Extend containment to instance segmentation mask suppression. Additional benchmarks: DeepFashion [31], Google Landmarks [32].

References

- [1] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan, "Object Detection with Discriminatively Trained Part-Based Models," *IEEE TPAMI*, vol. 32, no. 9, pp. 1627–1645, 2010.
- [2] N. Bodla, B. Singh, R. Chellappa, and L. S. Davis, "Soft-NMS — Improving Object Detection with One Line of Code," in *ICCV*, 2017, pp. 5561–5569.
- [3] J. Hosang, R. Benenson, and B. Schiele, "Learning Non-Maximum Suppression," in *CVPR*, 2017, pp. 4507–4515.
- [4] S. Liu, D. Huang, and Y. Wang, "Adaptive NMS: Refining Pedestrian Detection in a Crowd," in *CVPR*, 2019, pp. 6269–6278.

- [5] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU Loss: Faster and Better Learning for Bounding Box Regression," in *AAAI*, 2020, pp. 12993–13000.
- [6] Y. He, C. Zhu, J. Wang, M. Savvides, and X. Zhang, "Bounding Box Regression with Uncertainty for Accurate Object Detection," in *CVPR*, 2019, pp. 2888–2897.
- [7] X. Wang, T. Kong, C. Shen, Y. Jiang, and L. Li, "SOLO: A Simple Framework for Instance Segmentation," in *ECCV*, 2020. (Introduces Matrix NMS for parallel suppression.)
- [8] X. Chen et al., "Detectron2: A PyTorch-based Modular Object Detection Library," Meta AI, 2020.
- [9] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common Objects in Context," in *ECCV*, 2014, pp. 740–755.
- [10] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL Visual Object Classes (VOC) Challenge," *IJCV*, vol. 88, no. 2, pp. 303–338, 2010.
- [11] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal Speed and Accuracy of Object Detection," *arXiv:2004.10934*, 2020.
- [12] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-End Object Detection with Transformers," in *ECCV*, 2020, pp. 213–229.
- [13] S. Vicente, J. Carreira, and L. Agapito, "Object Detection by Box Voting," in *ICCV Workshop on Visual Domain Notes and Challenges*, 2015.
- [14] B. Jiang, R. Luo, J. Mao, T. Xiao, and Y. Jiang, "Acquisition of Localization Confidence for Accurate Object Detection," in *ECCV*, 2018, pp. 784–799.
- [15] L. Chu, Z. Liu, J. Wang, and Q. Tang, "Detection Crowded Object via Probabilistic Embedding Detector," *arXiv:2006.06626*, 2020.
- [16] R. Shi et al., "PP-ShiTU: A Practical Image Recognition System," *arXiv:2111.00760*, 2022.
- [17] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," in *NeurIPS*, 2015, pp. 91–99.
- [18] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *CVPR*, 2016, pp. 779–788.
- [19] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *CVPR*, 2016, pp. 770–778.
- [20] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature Pyramid Networks for Object Detection," in *CVPR*, 2017, pp. 2117–2125.
- [21] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object Detection via Region-Based Fully Convolutional Networks," in *NeurIPS*, 2016, pp. 379–387.
- [22] H. Oh Song, Y. Xiang, S. Jegelka, and S. Savarese, "Deep Metric Learning via Lifted Structured Loss," in *CVPR*, 2016, pp. 4004–4012.
- [23] J. Philbin, O. Chum, M. Isard, J. Sivic, and A. Zisserman, "Object Retrieval with Large Vocabularies and Fast Spatial Matching," in *CVPR*, 2007.
- [24] H. Hu, J. Gu, Z. Zhang, J. Dai, and Y. Wei, "Relation Networks for Object Detection," in *CVPR*, 2018, pp. 3588–3597.
- [25] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in *NeurIPS*, 2012, pp. 1097–1105.
- [26] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," in *ICML*, 2017, pp. 1321–1330.
- [27] J. Johnson, M. Douze, and H. Jégou, "Billion-Scale Similarity Search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [28] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," in *CVPR*, 2019, pp. 4690–4699.
- [29] A. Gordo, J. Almazán, J. Revaud, and D. Larlus, "Deep Image Retrieval: Learning Global Representations for Image Search," in *ECCV*, 2016, pp. 241–257.
- [30] F. Radenovic, G. Tolias, and O. Chum, "CNN Image Retrieval Learns from BoW: Unsupervised Fine-Tuning with Hard Examples," in *ICCV*, 2016, pp. 1212–1220.
- [31] Z. Liu, P. Luo, S. Qiu, X. Wang, and X. Tang, "DeepFashion: Landmark Detection and Detailed Category Classification," in *CVPR*, 2016, pp. 4675–4683.

[32] T. Weyand, A. Araujo, B. Cao, and J. Sim, "Google Landmarks Dataset v2 — A Large-Scale Benchmark for Instance-Level Recognition," *arXiv:2004.01804*, 2020.

[33] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized Intersection over Union: A Metric and A Loss for Bounding Box Regression," in *CVPR*, 2019, pp. 658–666.

Appendix A: Proofs

A.1. Formal Statement of the Containment Blind Spot

This appendix provides the full proof of Proposition 1 (Section 4.1), restated here in ε - δ form.

Theorem 1 (Restatement of Proposition 1). *For any $\varepsilon > 0$, there exists a containment configuration $b_i \preceq b_j$ such that $\text{IoU}(b_i, b_j) < \varepsilon$.*

Proof. Fix b_j with area A_j . Choose b_i with area $A_i = \frac{\varepsilon}{2} \cdot A_j$, positioned entirely inside b_j . Then:

$$\text{IoU}(b_i, b_j) = \frac{A_i}{A_j} = \frac{\varepsilon}{2} < \varepsilon$$

Since $\varepsilon/2 < \varepsilon$ for all $\varepsilon > 0$, the containment is invisible to any IoU threshold above $\varepsilon/2$. $\square$

Remark. The area ratio must be set to $\varepsilon/2$ (not ε) to ensure the *strict* inequality $\text{IoU} < \varepsilon$. Setting $A_i = \varepsilon \cdot A_j$ yields $\text{IoU} = \varepsilon$, which is equal—not strictly less than— ε .

A.2. Score Inheritance Propagation

Proposition 3 (Restated). *In a nested containment chain $b_1 \supseteq b_2 \supseteq \dots \supseteq b_k$, the outermost box b_1 receives final confidence $S[b_1] = \max_{1 \leq i \leq k} s_i$.*

Proof. By the area-descending processing order, boxes are visited as $b_1 \rightarrow b_2 \rightarrow \dots \rightarrow b_k$ (since $A_1 \geq A_2 \geq \dots \geq A_k$ by containment). b_1 enters the kept list K first.

Base case ($t = 2$): When b_2 is processed, it is contained in b_1 (Line 8). If the area-ratio gate passes ($A_2/A_1 < \alpha$), score inheritance executes $S[b_1] \leftarrow \max(S[b_1], s_2)$. After processing: $S[b_1] = \max(s_1, s_2)$.

Inductive step: Assume after processing b_{t-1} , $S[b_1] = \max_{1 \leq i < t} s_i$. When b_t is processed, it is contained in b_1 and score inheritance executes $S[b_1] \leftarrow \max(S[b_1], s_t) = \max(\max_{1 \leq i < t} s_i, s_t) = \max_{1 \leq i \leq t} s_i$.

At $t = k$: $S[b_1] = \max_{1 \leq i \leq k} s_i$. $\square$

Note: The area-ratio gate must pass for each containment in the chain. If $A_t/A_1 \geq \alpha$ for some t , the inheritance chain is broken at b_t (though b_t itself may still be contained by an intermediate box b_j with $A_t/A_j < \alpha$). In practice, nested containment chains with extreme area ratios are rare.

Appendix B: Algorithm Pseudocode

Algorithm: SANMS (Structure-Aware Post-NMS Refinement)

Input:

```
boxes[N][4]    --- NMS-filtered bounding boxes [x1, y1, x2, y2]
scores[N]      --- confidence scores
alpha          --- area-ratio gate (default: 0.75)
eps            --- float tolerance (default: 1e-3)
```

Output:

```
kept_boxes, kept_scores --- refined set sorted by confidence descending
```

```
1: areas = compute_areas(boxes)
2: order = argsort(areas, descending)
3: keep = []
4:
5: for i in order:
6:     is_contained = False
7:     for j in keep:
8:         // Containment check with tolerance
9:         if boxes[j].x1 <= boxes[i].x1 + eps and
10:            boxes[j].y1 <= boxes[i].y1 + eps and
11:            boxes[j].x2 >= boxes[i].x2 - eps and
12:            boxes[j].y2 >= boxes[i].y2 - eps:
```

```

13:
14:         // Area-ratio gate
15:         if areas[i] / areas[j] < alpha:
16:             // Score inheritance
17:             scores[j] = max(scores[j], scores[i])
18:             is_contained = True
19:             break
20:
21:     if not is_contained:
22:         keep.append(i)
23:
24: // Sort by final confidence descending
25: final_order = argsort(scores[keep], descending)
26: return boxes[keep][final_order], scores[keep][final_order]

```

Appendix C: Python Implementation

```

import numpy as np

def sanms(bboxes: np.ndarray, scores: np.ndarray,
          alpha: float = 0.75, eps: float = 1e-3) -> tuple:
    """
    Structure-Aware Non-Maximum Suppression (SANMS)
    -- Post-NMS Refinement Module

    Resolves geometric containment relationships left unhandled by
    IOU-based NMS. Designed for subject detection in visual retrieval.

    Args:
        bboxes: (N, 4) array, NMS-filtered boxes [x1, y1, x2, y2]
        scores: (N,) array, confidence scores
        alpha: area-ratio gate (default 0.75)
        eps: float tolerance for containment check (default 1e-3)

    Returns:
        kept_bboxes, kept_scores (sorted by confidence descending)
    """
    if len(bboxes) <= 1:
        return bboxes.copy(), scores.copy()

    areas = (bboxes[:, 2] - bboxes[:, 0]) * (bboxes[:, 3] - bboxes[:, 1])
    order = np.argsort(areas)[::-1] # area descending

    keep = []
    scores = scores.copy().astype(np.float64)

    for i in order:
        is_contained = False
        for j in keep:
            # Containment check with tolerance
            if (bboxes[j, 0] <= bboxes[i, 0] + eps and
                bboxes[j, 1] <= bboxes[i, 1] + eps and
                bboxes[j, 2] >= bboxes[i, 2] - eps and
                bboxes[j, 3] >= bboxes[i, 3] - eps):

                # Area-ratio gate
                if areas[i] / areas[j] < alpha:
                    # Score inheritance
                    scores[j] = max(scores[j], scores[i])
                    is_contained = True
                    break

        if not is_contained:
            keep.append(i)

    final_order = np.argsort(scores[keep])[::-1]
    return bboxes[keep][final_order], scores[keep][final_order]

```

Appendix D: Supplementary Details

D.1. Synthetic Data Generation Protocol

The synthetic data generation produces controlled detection outputs with known containment patterns:

1. **Subject box generation:** For each image, a ground-truth subject box is randomly placed. A container box (loose crop, +10--20% margin) and a contained box (tight crop, 20--40% of subject area) are generated around the subject.
2. **Confidence assignment:** The contained box receives high confidence ($s \sim \mathcal{N}(0.9, 0.05)$) because local textures are distinctive. The container box receives lower confidence ($s \sim \mathcal{N}(0.6, 0.05)$) because the detector is less certain about the full extent.
3. **Feature modeling:** The contained box's crop yields poor retrieval features (noise $\sigma = 0.6$) because it captures only a fragment. The container box's crop yields good retrieval features (noise $\sigma = 0.15$) because it captures the complete object.
4. **Noise injection:** Additional non-contained boxes (random placements, partial overlaps) are generated to simulate realistic detector output.
5. **Parameters:** 200 images per run, 3 independent random seeds. Containment area ratio enforced at < 0.6 .

D.2. Box Voting Implementation

Box Voting [13] is implemented with $\tau_{\text{vote}} = 0.5$. The method averages coordinates of boxes whose pairwise IoU exceeds the threshold. Since contained boxes have near-zero IoU with their containers ($\text{IoU} = A_{\text{small}} / A_{\text{large}} \ll 0.5$), they are excluded from the voting pool. This is confirmed empirically: Box Voting produces identical output to the NMS baseline (Table 5).

D.3. Statistical Analysis

All results are reported as mean $\pm$ standard deviation over 3 independent runs with different random seeds. The controlled setup ensures that performance differences arise solely from the post-processing algorithm, as all methods receive identical detector output.